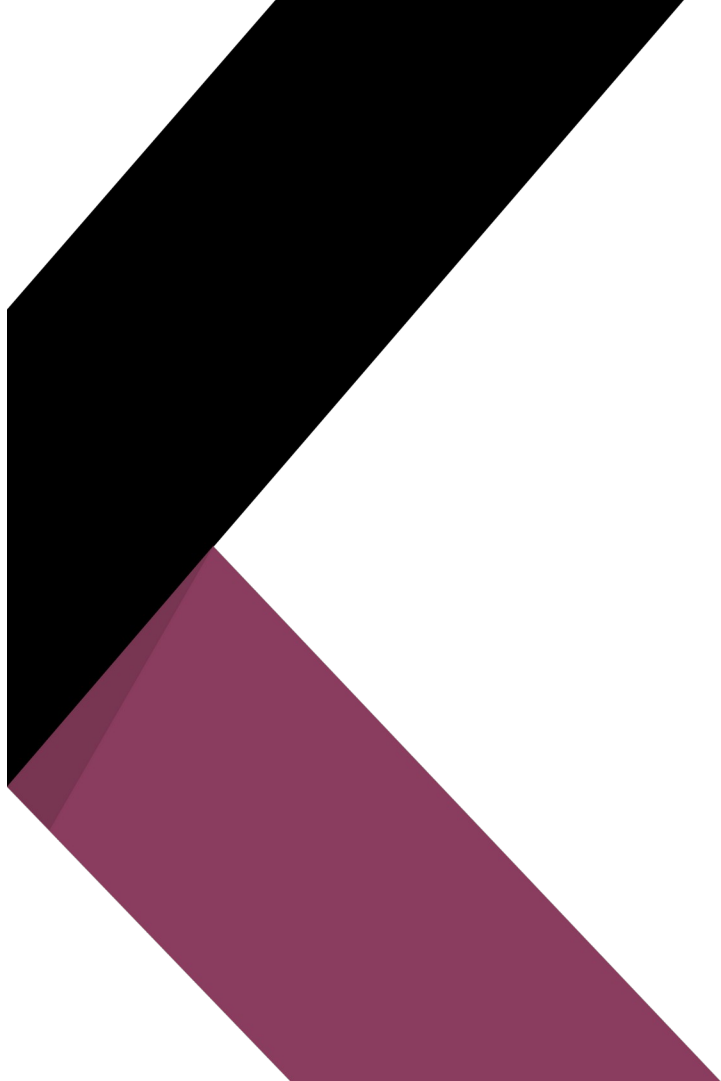






Braking System

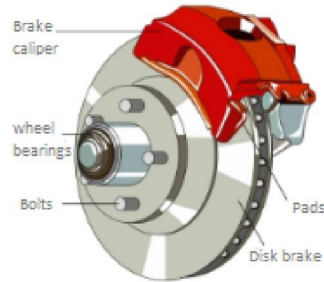


Types of brakes

Definitions

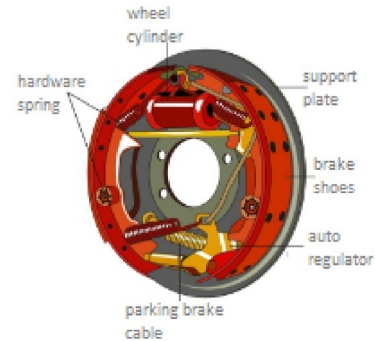
Disk brake

Positioned inside the wheel, operated by a hydraulic system that uses the variation of the liquid inside it to impart to the pads a force proportional to that exerted on the pedal.



Drum brake

It consists of a metal rotating cylinder (drum) and one or more surfaces coated with friction material (shoes). When the driver is braking, the shoes are pushed towards the drum creating a frictional force capable of slowing or braking the vehicle.





Oversteer and understeer

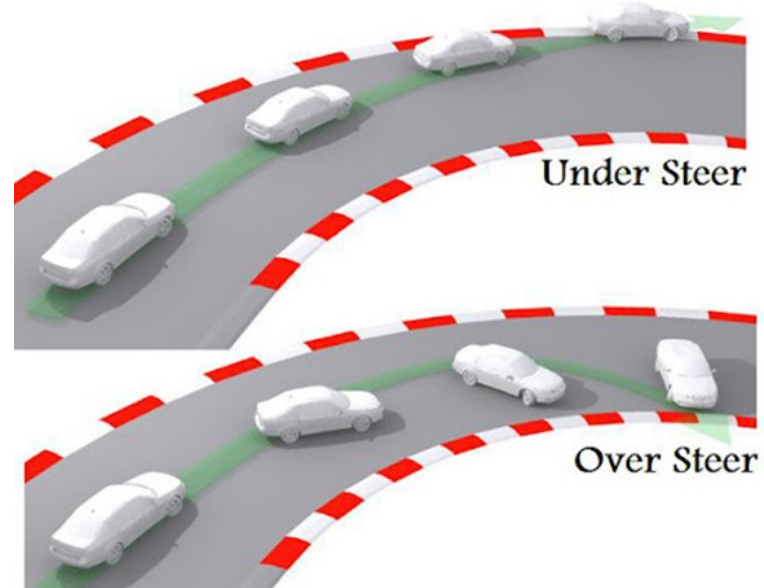
Definitions

Oversteer

Behavior of a car that during curved trajectory tends to travel a tighter trajectory than the one intended by the driver. If a sudden loss of adhesion of the rear axle happens, there is the phenomenon of spinning or skidding.

Understeer

Behavior of a car during a curve that tends to widen the trajectory of travel, forcing the pilot to make correction maneuvers (leave the gas and realign the steering for a moment) to stay on the desired trajectory. It is a behavior that occurs more on front-wheel drive cars when they go through a curve at speed or in wheel lock case.





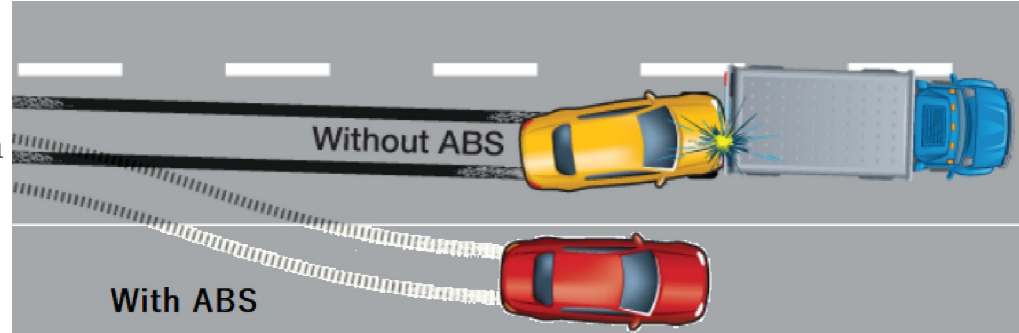
Vehicle Subsystems

Antilock Brake System



On each wheel of the vehicle there is a sensor (angular position transducer + photo sensor) and an inductive proximity sensor. The ECU measures the speed of rotation of the wheel and if it detects the locking of one or more wheels, it acts to inform the driver.

ABS

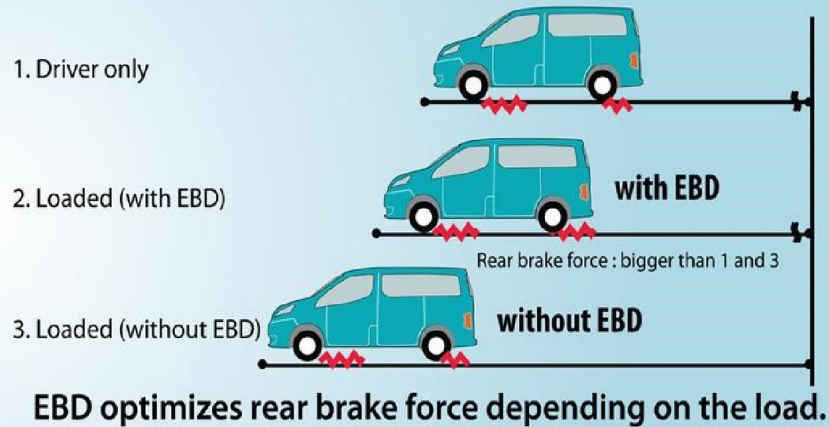




Vehicle Subsystems

EBD

Electronic Brakeforce Distribution



During braking, the braking load on the different wheels of the car is not the same. This can cause a possible locking of the rear wheels which, being more flat, and have less grip with the asphalt. This could lead to a complete lack of directionality of the car, that can spin. The brake distributor avoids this problem, especially when cornering, by lightening the braking force on one or both rear wheels avoiding locking.



Vehicle Subsystems

ESC

Electronic Stability Control



It is also known with the BOSCH commercial name **ESP** (Electronic Stability Program) or

It is a system for stability control of vehicles which acts when skidding happens, adjusting engine power and braking the individual wheels with different intensity in such a way as to stabilize the car's attitude.

The control unit intervenes both on the engine supply of the motor (reducing its torque) and on the individual calipers brake, correcting the dynamics of the car.

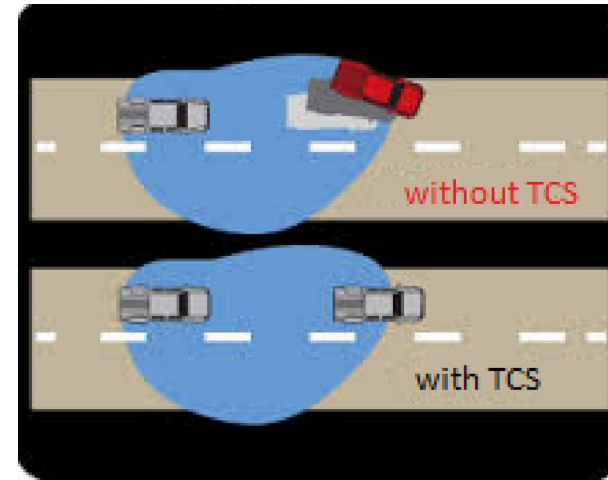




Traction Control System (Anti Spin Regulation)

It is an electronically managed system that prevents the drive wheels of a vehicle from spinning during acceleration. This system is useful in case of critical conditions (rain / ice), more generally in case of loss of traction of a wheel. The system identifies the wheel spinning through sensors placed on all wheels and thanks to an electronic processor which, processing the acquired data, intervenes on some variables, which are different according to the type of control of the system:

- Systems that acts on the brake
- Systems that acts on the engine power supply



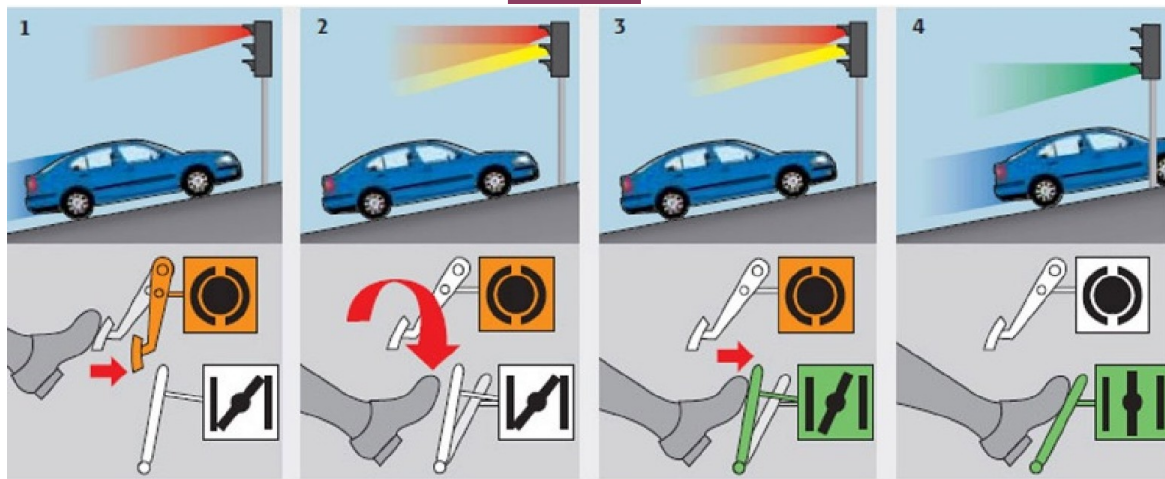
Vehicle Subsystems

HH

Hill Holder



In the case of starting from a standstill on a hill, it prevents the vehicle from rolling backwards, in the time that passes between the release of the brake and the actual start.





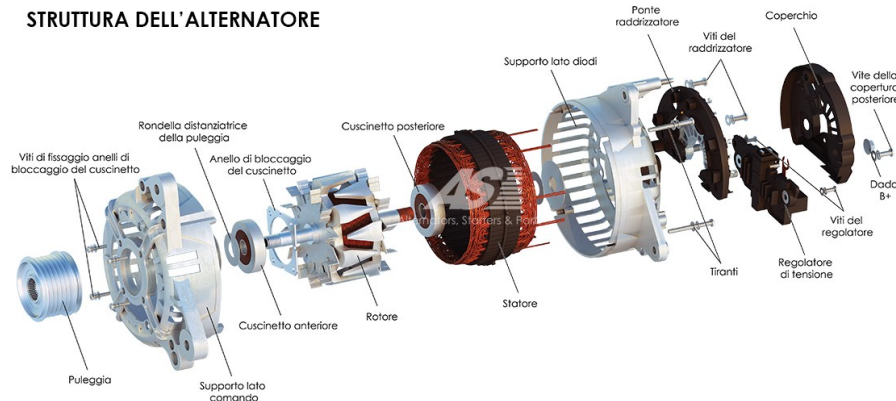
Alternators



The alternator has the function of keeping the battery charged, which is necessary to start the engine and to power all the electrical functions on board. Since there is no way to store electricity in alternating current, there is a rectifier bridge that has the function of transforming it into direct current and thus allowing its accumulation.

In the automotive field, the concept of SAM (Smart Alternator) was introduced, which differs from the classic alternator because it is able to vary the power subtracted from the engine according to the running state of the vehicle.

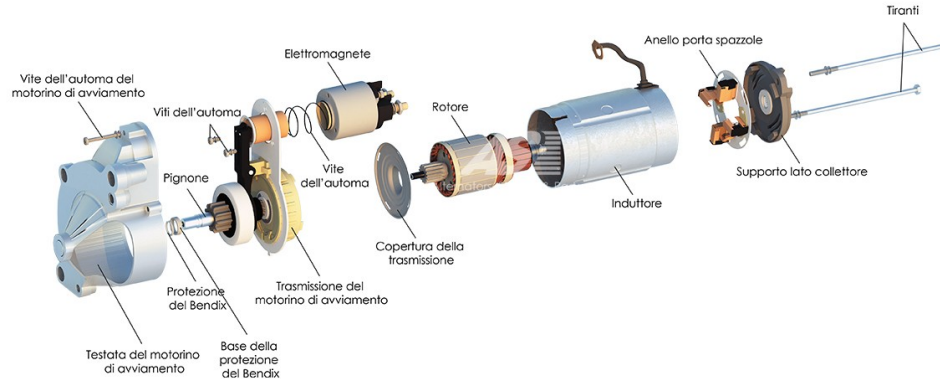
STRUTTURA DELL'ALTERNATORE





The starter motor is a component of the internal combustion engine, integrated into the starting system of the engine, which allows the engine to reach a number of revolutions sufficient for self-sustaining. It absorbs current from the battery to rotate the engine, and it is mechanically connected on the same shaft.

STRUTTURA MOTORINO D'AVVIAMENTO





Emission related subsystems

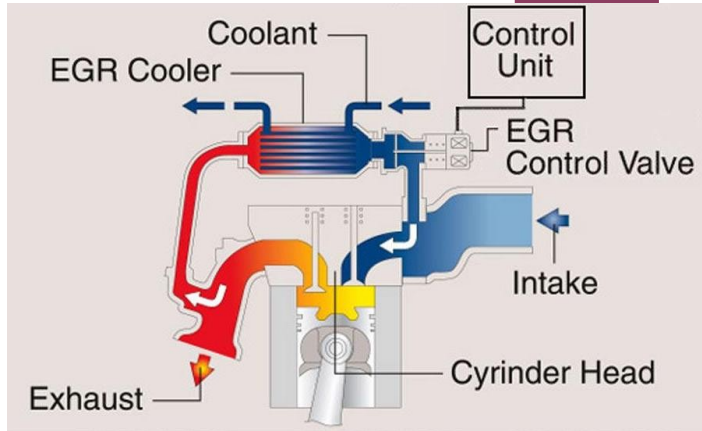


Vehicle subsystems

EGR

Exhaust Gas Recirculation

The recirculation of exhaust gases or recirculation of fuel gases (in the field of alternative internal combustion engines) consists in recirculating a "small" part (5-15%) of the exhaust gases by making them pass through the exhaust manifold to the intake manifold, in order to reduce a part of the pollutants present in the exhaust gases



This is controlled by the engine control unit via a PWM signal, allowing the regulation of the quantity of exhaust gases which are recirculated in the intake manifolds and aspirated into the engine.

The recirculation of burnt gases, inert gases and gases that do not participate in combustion, by decreasing the quantity of mixture useful for combustion and the power supplied, is made to lower the maximum temperatures of the cycle, thus reducing the emissions of oxides of nitrogen (NO_x).



Vehicle subsystem

DPF

Diesel Particulate Filter

A diagnosis and management software continuously monitors the filter to ensure correct operation and maintenance of the same. The filter in question is combined with a pre-catalyst and has the purpose of physically filtering fine dust. The system collects the combustible gases in the exhaust manifold and then conveys them to the catalytic converter, passing through the actual filter, proceeding towards the expansion tank, the silencer and the outlet.





The lambda probe is able to detect the oxygen concentration inside the exhaust gases. Through the comparison measurement of the oxygen present in the ambient air, measured at the other end of the probe, the quantity of air in the exhaust gases is indirectly obtained.

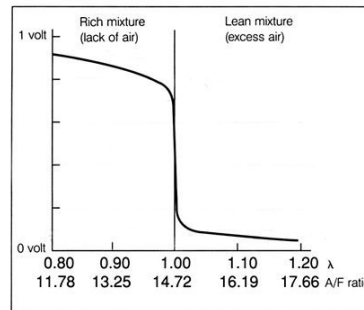
Lambda indicates the air / fuel ratio

$\lambda = 1$ stoichiometric combustion

$\lambda < 1$ fuel excess

$\lambda > 1$ air excess

The probe then transmits as an electrical signal to the control unit which regulates the introduction of fuel and air into the combustion chamber.





Types of lambda probes

- **Lambda probe with zirconium dioxide**

The outer surface of the zirconium dioxide element is in direct contact with the exhaust gases, while the inner surface is with the atmosphere.

- **Lambda probe with titanium dioxide**

The titanium dioxide element does not generate a tension like the zirconium. By applying a suitable voltage value to the titanium probe, a current can be measured in output, which is related to the oxygen concentration in the exhaust gases.

- **Double lambda probe**

The single lambda probe sends a two-stage signal (presence or absence of oxygen): with the use of only one sensor, the control unit (ECU) must constantly work on the correction of the injection values e ignition. The use of two probes allows to have a more truthful sampling and to increase the duration of the sustem, as sensors tend to vary their sensitivity with aging.





Vehicle Subsystems

Catalytic converter

The catalytic converter, known as a catalytic converter or also a catalyst, is a device mounted on the exhaust system of an internal combustion engine, which operates preferably between 180 and 380 ° C. This element acts as a suppressor of harmful gas emissions, promoting complete oxidation and reduction of exhaust gases.

It also contributes together with the silencer to reduce the exhaust noise.





Types of catalyst

- Reducing

It is also called one-way because it acts on the main pollutant. It is used for diesel engines: a rhodium base (Rh), able to reduce nitrogen oxides (NOx) into oxygen and nitrogen.

- Oxidant

It is also called Bidirectional, as it acts on the two main pollutants. It is used for engines with controlled ignition, based on Platinum and / or Palladium, that is capable of oxidizing carbon monoxide (CO) and unburnt hydrocarbons (HxCx)

- Trivalent or oxidizing and reducing

It is also called three-way, as it acts on the three main pollutants. It is characterized by a first reducing catalyst and a second oxidizing catalyst, is used in petrol and gas engines with stoichiometric combustion and equipped with control lambda.



Vehicle Subsystems

SCR

Selective Catalyst Reduction

Selective catalytic reduction is a chemical process for the reduction of NO_x from exhaust gases.

SCR devices are commonly used in both industrial combustion and automotive applications (Diesel vehicles).

A liquid or gaseous reducing chemical agent (usually ammonia or urea) is added to the exhaust gas in the presence of a catalyst. The reductant has a strong tendency to absorb oxygen thus limiting the formation of NO_x within the exhaust gas.

A fundamental point of the SCR process is the catalyst: this must be able to make NO react with O₂ and not NH₃.



Valve Timing



The timing is intended as the optimum opening point of the various valves of a four-stroke engine and is obtained by synchronizing the camshaft to the crankshaft.

This synchrony is obtained with a chain, belt or with gears.

The timing can be fixed or variable.

Fixed Timing

This type of timing is characterized by a simple type camshaft that has a height fixed, therefore it does not decrease the various phases of the cycle, and it gives a constant phasing.

It does not postpone or anticipate neither the opening nor the closing of the valves according to the speed, so with this type of timing the engine has a classic distribution



Variable Timing

A variable valve timing engine has a distribution pattern that varies with varying engine speeds, generally aids or special camshafts are used which under certain conditions, or modify the timing or the number of valves that work, allowing to optimize the performance and consumption (thus also reducing pollution).

Another reason why you can use this scheme is because it allows a differentiated behavior of the engine, so that up to a certain number of rpm or electronic setting, there is a "smooth" engine behavior.



Types of control

- Camshaft rotation

With this type of control it is possible to move the timing without reduce its amplitude.

- Double toothed gear

In this case is used a gear interposed inside the gear which is moved by the distribution and has the ability to slide on the distribution shaft.

- Variation of cam effect

The cam acts on the poppet valve and therefore is able to vary the amplitude of the valve timing and on the opening of the valves.



- Double cam

Several cams are used to actuate the valves on the same side. These cams have different profiles, and depending on the situation, thanks to a system interposed between the lobes and the valves, you will have a less pronounced or more pronounced actuation by the cam

- Rocker arm

Only one cam is used, but a rocker arm is used to actuate the valves. The rocker arm is regulated by a second lobe system, which allows the increase or decrease of the effect of the lobes of the distribution system.

- Disengageable cams

The cams that are used can slide along the camshaft, thanks to a piloted system, and allowing the variation of the timing or the locking in closed position of the valves, causing the engine to run with fewer cylinders.



Variable Timing Systems

There are various timing systems, with different actions from each other

- Alfa Romeo phase variator is a valve timing variation system designed by Alfa Romeo, the first used in a series production car
- VVT- and VVT-iE, are systems designed by Toyota
- VTEC, is a type of system designed by Honda and produced in many variants, including 3-stage VTEC
- MIVEC, is a type of system designed by Mitsubishi
- VarioCam and VarioCam Plus, are systems designed by Porsche
- VVL, N-VCT, CVTC and VVEL are systems designed by Nissan
- VANOS and Valvetronic, are systems designed by BMW
- AVCS and AV, are systems designed by Subaru
- S-S-V, is a type of system designed by Mazda
- VCT, is a type of system designed by Ford
- VVC, is a type of system designed by Rover
- VFD is a system of valve timing variation designed by FIAT, such as the Multiair, which allows of any parameter of the opening of the cams.
- CamTronic is a system of some valves from Daimler AG
- Shut off is a system that allows you to not operate by sliding the cams along the shaft itself, permitting



Multiair

Multiair is a valve opening control system in petrol engines and diesel engines, created by FIAT in 2002, and presented for the first time on a series engine (a FIRE engine) in 2009.

The aim of the designers of the Multiair is to provide, in a cam system, the same flexibility as camless systems.

In a traditional engine, the cam allows you to assign the valve opening and closing profile. There opening / closing law is fixed and can be optimized through a variable valve timing only for a restricted operating range, discriminated by the RPM.

In the Multiair system, the tappet consists of two rigid elements connected by means of a pumping element containing oil. The pumping element is equipped with a valve controlled by a dedicated electronic system.

If initially the valve of the pumping element is closed, the tappet will follow the opening profile imposed by the cam. By opening the pumping element valve at a suitable time (according to the operating situations and when the throttle control is activated), the tappet will be recalled by the spring and, consequently, will not come reached maximum opening or the closing of the same will be anticipated. If in the meantime it is requested more power, the control valve will close again, preventing more oil from escaping, allowing you to use (if still available) the remaining profile of the cam, but without being able to regain the travel lost with oil leaking.



Multiair

How does it work?





Transmission



The gearbox is a mechanical component that has the function of modifying the characteristic of the torque (and not of the power) coming out of an engine, similar to a speed reducer. It also allows you to select a different gear ratio from time to time, depending on the range the gearbox is equipped with.

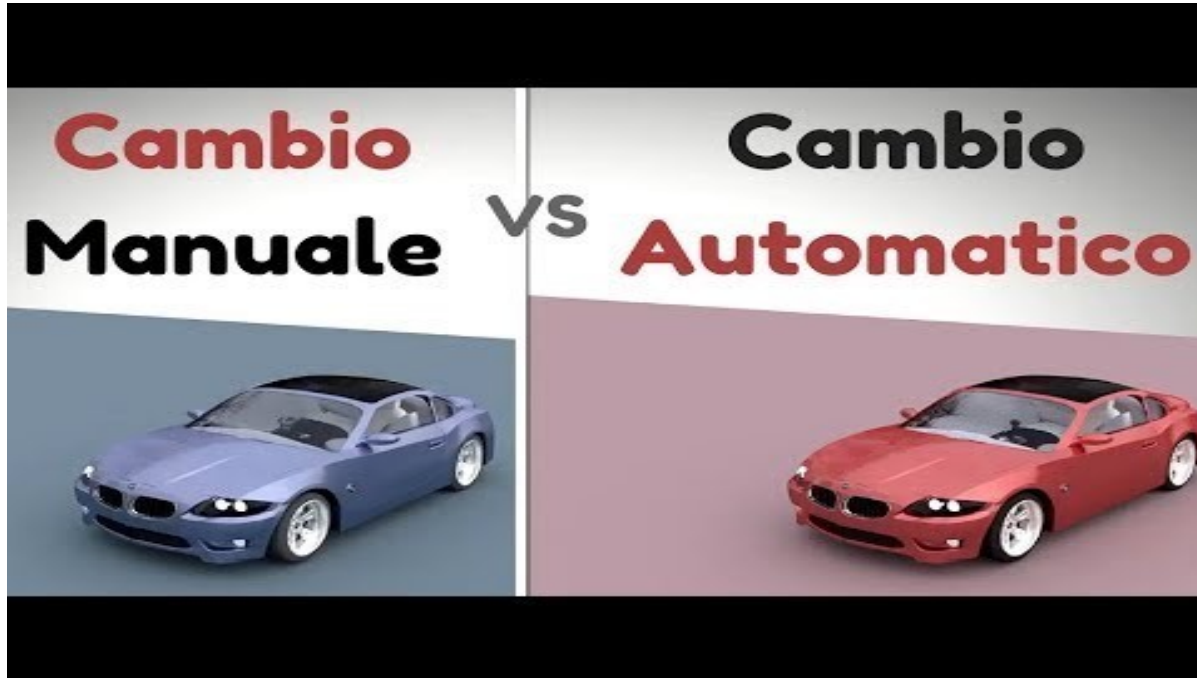


The gearbox allows to vary the ratio between engine speed and vehicle speed, in order to obtain an appropriate drive torque for the wheels. In fact, the engine has different optimal rotation speeds depending on whether it is necessary to favor, if mileage efficiency (generally at constant speed), or torque and power (for maximum speed, pickup, acceleration, climbs, etc.) or under torque (for snow-covered or low-grip surfaces).



Transmission

Manual transmission vs Automatic transmission





Transmission

MTA

Manual Transmission Automated

The robotic gearbox is a type of gearbox for automotive use, also called Robotized Manual Gearbox and derives from the "robotization", replacement of manual drives with automatic drives, of a normal manual gearbox. By extension, even gearboxes originally designed to be operated by automatic actuators but with mechanics similar to manual gearboxes, therefore with gears that can be coupled with sleeves and synchronizers, are also considered robotized.

The electronic control unit, by means of suitable actuators, provides for the clutch movements and the selection / engagement of the gears, autonomously. This control unit carries out the process of detaching the clutch, the passage of the transmission ratio and the subsequent refitting of the clutch.



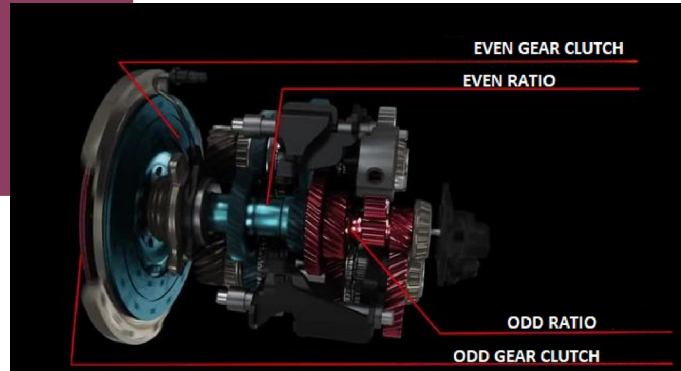
Transmission

DCT

Dual Clutch Transmission

It is a gearbox in which there are two shafts connected to two clutches in turn, connected to the shaft transmission. On one shaft there are odd ratios while on the second there are even ratios: what happens is a simultaneous rotation of the internal shafts, however only one of the two, thanks to a clutch, transfers the motion to the transmission shaft. Meanwhile, the other gear ratio shaft continues to rotate and has the next ratio "ready".

The advantage consists in a remarkable shifting speed.





Transmission

AT

Automatic Transmission

The most common type of automatic gearbox is that with epicycloidal rotism. A group of epicyclic systems in series allow each one to make a gear change by means of the intervention of brakes which act, depending on the case, on the crown, pinion or satellite carrier.

Automatic shifting is usually achieved by exploiting the pressure of a fluid in a hydraulic circuit essentially operated by a final speed detector.

A transmission with hydraulic epicycloidal rotism gearbox usually consists of the following parts:

- Torque converter
- Epicycloidal rotism gear
- Actuator



Transmission

AT

- Torque converter

It is a device interposed between the engine and the gearbox which replaces the traditional clutch. It produces a variation in torque at low speeds and during accelerations, but above all, it operates "at idle" (at minimum engine speed) and it doesn't transmit torque, if not of a minimum entity and in any case always conquerable by a minimum brake action.





Traction



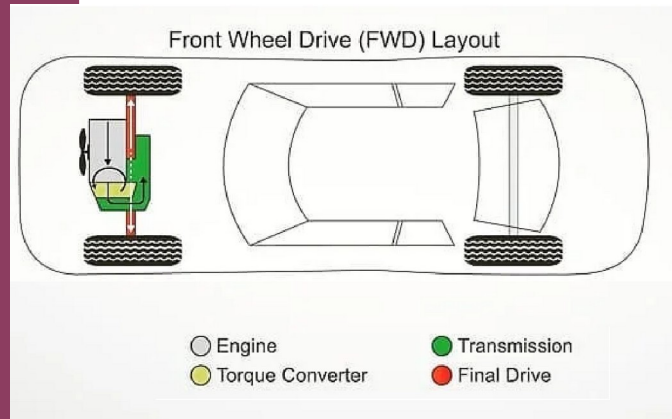
Traction

FWD

Front Wheel Drive

Front-wheel drive cars have front wheels that transmit motion.

Front-wheel drive cars can also be divided according to the arrangement of the engine with respect to the driving wheels. In fact they can be with front motor (the motor is mainly arranged in front of the axis of the wheels) or set back (the engine is mostly located within the wheelbase of the vehicle).

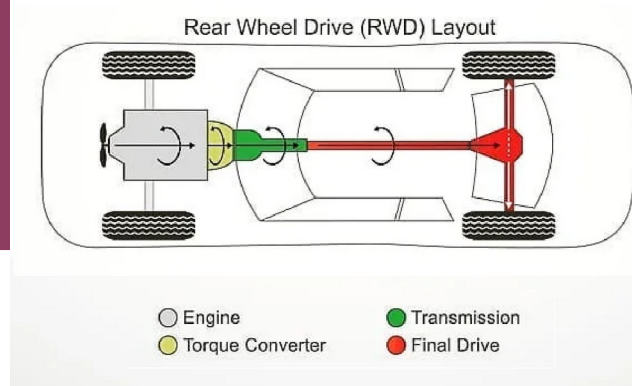




Traction RWD

Rear Wheel Drive

A rear-wheel-drive arrangement needs the engine and the transmission to be mounted longitudinally. The engine torque is transferred to the rear wheels. Usually, sports cars with powerful engines have RWD.



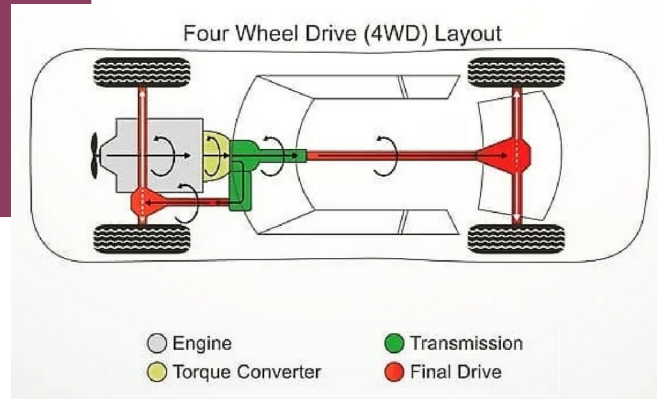


Traction

4WD

4 Wheel Drive

The vehicle's engine torque is distributed equally on all wheels. In these cases it can also be called 4×4 . The traction is all-wheel, which is versatile for traveling on uneven surfaces and for giving the necessary stability to vehicles, which can be off-road vehicles or cars designed for those who want to use their vehicle not only to go on the concrete road.





Traction

AWD

All Wheel Drive

The traction system is capable of distributing the drive torque delivered by the engine to both axles and to all the wheels. The driver does not have to do anything to activate the system. The power is distributed over all the wheels or the ECU automatically assigns the torque to maintain the traction.

The main difference with the 4WD is that a 4WD transmission sends equivalent power to both the front and rear axles. An AWD shifts the power between the axles as required.

The AWD is suitable for everyday conditions, is used on some off-road vehicles and for some high-performance riding. The 4WD is suitable for heavier tack.

CONTACTS

Via G. Porzio,
Centro Direzionale Isola B3
80143 Napoli Via G. Porzio

www.kineton.it

